Post-Disaster Damage Assessment and Rescue Guidance

CSE499B – Section 13, Group 5, Update 3

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Weekly Update:

Our system reads before-and-after satellite images of a disaster, marks the damage building by building, counts it, and is to write a rescue report that cites official guidelines, offline, on hardware small enough for the field. Last week's tests showed that our segmentation model fails mainly at *finding* rubble, not at grading it. This week we report what that led to, a find-then-grade model: SAM 3, Meta's general segmentation model, is given the word "building" and outlines every building on the before-disaster image, where buildings still stand; then a ResNet-18 image classifier that we trained compares a crop of each building from the before-image with the same crop from the after-image, and labels it intact, damaged, destroyed or not a building. It is the first model to reach our destroyed-building target on clean validation data, narrowly. We also report our vision-language model (VLM: a fine-tuned Qwen2.5-VL-7B that answers questions about the images), kept after 28 more training steps on full-size images, and two tests by Aryan and Tamanna showing that the VLM should neither count nor grade buildings itself. Segmentation scores are on our 645 validation image pairs, or, where marked clean, on the 605 left after removing 40 whose before-image also appears in training.

Results Summary

Goal	Target	Best result so far	Status
1. Damage map	Mean IoU above 0.52 and destroyed-building IoU above 0.30, on two training seeds	Find-then-grade, clean pairs: destroyed-building IoU 0.316, 0.305 and 0.298 with three graders (mean 0.307); mean IoU 0.547	Met on the mean of three graders, narrowly
2. Counting	Above the base VLM's 34.2% on the benchmark's counting questions	32.96% by a fixed query over a table of the buildings our U-Net found (32.47% without duplicated images)	1.7 points short
3. Offline report	The report-writing agents on our own models, with damage map, counts and rescue heatmap	Map, counts and heatmap done; guideline search passes its test (27 of 30); agents moved onto our 7B	Offline proof run next
4. Field model	A small segmenter and a distilled 3B VLM	U-Net in 8-bit integers: 33 MB, 1.43× faster on a CPU, destroyed IoU down 0.0003; 1,146 training answers for the 3B	Segmenter done; 3B training next

IoU (intersection over union): the overlap between the pixels predicted as a label and the pixels truly of it, from 0 to 1; mean IoU averages it over background, intact, damaged and destroyed.

Abrar Mohammed Tanzim Alam – Find-then-Grade, VLM and Field Model

1. **The U-Net that reads both images (R11):** our U-Net given the before-image as three extra input channels. It raised mean IoU from 0.544 to 0.588 (all 645 pairs, averaging 8 flipped and rotated copies of each image), our best, but not the destroyed-building score; on clean pairs that falls to 0.269, below the 0.287 of R9 (our best U-Net that sees only the after-image), because the duplicated pairs share exactly the before-image R11 now reads. R12, R11 retrained with 1,619 extra xBD images, changed nothing. Turkey's 2023 earthquake holds 37% of all destroyed pixels and R11 scores 0.000 there; without Turkey it reaches 0.410.

2. **Find-then-grade (R16c), how it works:** SAM 3 runs on the before-image cut into four 512 px quarters, keeping outlines it is at least 50% confident in. Each outline is cropped from both images at 128 px with a 32 px margin, and a Siamese ResNet-18 (one network reads both crops, and a final layer compares them) outputs intact, damaged, destroyed or "none" (mostly background). It was trained on SAM 3's outlines of 5,218 training images, each labelled with the main true grade beneath it (or "none"); its stopping epoch and a bias per grade, which corrects for intact buildings being far more common, were chosen on 580 held-out training images, never on validation. Each building's grade is painted into its outline to make the damage map.
3. **Find-then-grade, results:** with perfect grades on SAM 3's buildings the destroyed-building IoU would be 0.556, so most of the gap is the grader. Ours reached 0.335 and 0.324 on all 645 pairs, and 0.316, 0.305 and 0.298 on clean pairs with three graders trained identically but for their random seed (mean 0.307): the target is met on the mean, narrowly. Caveats: this map (SAM 3's buildings only, without R11's extra pieces) was singled out after its scores were seen, and its mean IoU (0.547) is below R11's. On the true validation buildings, the third grader is right on 94%, 67% and 70% of the intact, damaged and destroyed ones it grades.

4. **The duplicate check, extended to the benchmark:** we confirmed the 40 pairs of Ridita's audit by hashing the pixels of all 20,988 training and 7,278 benchmark images, re-scored every result without them, and found that 186 of the benchmark's 10,675 photo questions use an after-image identical to one in the instruction data our VLM was fine-tuned on; without them no score moves by more than 0.7 points.
5. **The VLM at full image size:** re-scored on all six benchmark tasks at the full image size used in the DisasterM3 paper (1036 px), then trained for 28 more steps at that size: LoRA adapters (small trainable matrices added to the frozen model), batches of 256, on 7,018 examples (1,627 disaster-type and 1,882 relational-reasoning questions, plus 3,509 from the other four tasks so

they are not forgotten). The rule fixed before the run: keep it only if disaster-type and relational-reasoning accuracy rise and no other task drops by more than 1 point. Relational reasoning rose 3.5 points ($p < 0.001$), disaster type 1.1, the other four moved by -0.95 to $+0.64$; mean 38.4% $\rightarrow$ 39.2%, the same without the 186 questions. Kept; its counting (30.3%) is still below 34.2%.

6. **Counting:** a SQLite table with one row per building R11 found in the benchmark's 2,652 test images (117,687 buildings: image, grade, position, size) answers each counting question with a fixed query: 32.96%.

7. **Field model and heatmap:** R9 exported to ONNX and stored in 8-bit integers, with its first, last and two full-resolution decoder layers kept in 32-bit: 33 MB, 1.6 s per image on a CPU (1.43× as fast as the 32-bit version), destroyed IoU down 0.0003. The rescue heatmap blurs R9's damaged and destroyed pixels into a danger map and circles each image's top three hot spots; on all 645 images, 73.6% of the circled points lie within 20 px of a truly damaged or destroyed building.

8. **Training data for the 3B (distillation: a small model trained to copy the 7B):** the new 7B answered 3,265 training questions with a short reasoning; the 1,146 whose final answer matches the label are kept (the old 7B gave 1,085).

Ridita Afrin Riya – Offline Guideline Search and Report Agents

1. **Retrieval test:** 30 questions, the guideline sections that should answer each, and a pass mark of 24 were fixed before any search ran. The search (all-MiniLM-L6-v2 sentence embeddings for meaning plus BM25 for keywords, merged by reciprocal rank fusion) returned an expected section in its top 3 for 27; meaning alone found 20. A committed Kaggle run with the internet switched off matched all 16 check searches.
2. **Report agents on our own model:** the four DisasTeller agents and their three image tools now send every request to our new 7B, served by vLLM on the same machine (any address but `localhost` is refused), instead of Google's online Gemini. A network log records every connection during a run, and five pass rules were fixed before any run, among them six reports and no connection outside the machine. A laptop run against a stand-in model server passed all five; the first Kaggle run loaded the 7B on two T4 GPUs and stopped on a file path, now fixed. The internet-off proof run is next.

Aryan Sami – Counts in the VLM's Prompt

1. **R11's counts in the prompt:** for 1,701 multiple-choice questions asking how many damaged or destroyed buildings an image has, R11's count and the buildings' boxes were written into the question. The fine-tuned 7B then scores 27.0%, against 15.6% from the image alone; but picking the option nearest R11's count, with no VLM, scores 37.0% on the same questions, and on 66 questions shared with Noman's test the SQL table beats the prompt 26 to 20. Counts should go from the table straight to the answer.

Tamanna Akter Mou – Can the VLM Grade a Building?

1. **Test against ground truth:** 300 true validation buildings (100 intact, 100 damaged, 100 destroyed), each cut out as a 256 px window, enlarged 2× and with a red box around the building. Our grader was right on 212, the new fine-tuned 7B on 100 and the base 7B on 99 (exact McNemar test on the buildings where only one of the two is right: $p < 10^{-18}$ for both). The fine-tuned 7B called 298 of the 300 intact, and the base 7B never said destroyed. Grading stays with the trained grader; the VLM's job in the report is to describe what the grader and the table found.

Next Week

- **Field model:** train the 3B on the 1,146 answers (whether it learns the answers alone or the reasoning too is still to be chosen).
- **Offline report:** the report agents' proof run on our 7B with the internet switched off, then a comparison of its six reports with the earlier ones written by Gemini.
- **Counting:** fixed queries over a table of the find-then-grade buildings, to close the 1.7-point gap.
- **Data:** split all future training and validation sets by before-image, so duplicates cannot cross the split again.